Andrew (Li-Jia) Chang

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Hamilton/Auckland, New Zealand

EDUCATION

University of Auckland
Bachelor of Engineering (Honours)

- Specialising in Computer Systems Engineering (Part III)

Project Experience

Data Management Desktop App — Python, Visual Studio Code, Matplotlib, GitHub, PyQt6
2026

- Developed a desktop application for importing and analysing CSV based financial datasets
- Designed an interactive graphical user interface for data visualisation and reporting
- Implemented forecasting algorithms to predict future financial trends using historical data
- Generated financial graphs and charts using Matplotlib to visualise performance across different periods
- Applied data processing and analysis techniques to support business decision making
- Maintained source code and documentation through GitHub

Energy Monitor Design — C, (Atmel) Microchip Studio, LTSpice, Proteus, Altium Designer, PuTTY, GitHub
2026

- Developed an ATmega328P based energy monitor to measure and display voltage, current, average power, and energy
- Implemented modular embedded C firmware for ADC sampling, signal processing, LCD output, timers, and UART communication
- Designed and simulated analogue measurement circuits using LTspice and Proteus, achieving measurement accuracy within project specifications
- Created a double-layer PCB in Altium Designer and managed team collaboration and version control using GitHub

Android PC Part Browsing App — Java, Kotlin, XML, Android Studio, Firebase, GitHub
2026

- Developed an Android application for browsing and comparing PC hardware components
- Implemented category based filtering, search functionality, top pick recommendations, and related item suggestions
- Integrated Firebase as a cloud database for storing and retrieving product information
- Created a wish list system allowing users to save and manage preferred components
- Developed a component comparison feature to evaluate hardware specifications side by side.
- Implemented a compatibility checker to verify compatibility between PC components and provide detailed compatibility feedback

Country Continent Calculator — Java, Visual Studio Code, Maven (MVN Wrapper), GitHub
2025

- Developed a desktop application that calculates the shortest travel distance and fuel efficient route between countries
- Implemented route finding algorithms to determine optimal paths between locations
- Applied software engineering best practices, including unit testing and structured project architecture
- Managed version control and collaboration using GitHub

Technical Skills

- **Programming and web technologies:** Java, Python, C, C++, C#, HTML, CSS, JavaScript, R, MATLAB
- **Data analysis and visualisation:** Microsoft Excel, Python (Matplotlib), R (RStudio), MATLAB and SQL; academic experience in dataset checking, statistical analysis, forecasting and reporting
- **Backend and databases (academic project experience):** Spring Boot, REST APIs, JPA/Hibernate, Spring Data JPA, SQL/JPQL, H2, Firebase, relational data modelling, JSON/Jackson, WebSockets/STOMP, API pagination and content negotiation
- **Networking and authentication (academic project experience):** HTTP, TCP socket programming, Java object serialization, JWT authentication, cookie handling and API authorization
- **Software engineering:** Unit and integration testing using JUnit 5 and MockMvc, dependency injection, object-oriented design, software architecture, database transactions and concurrency control
- **Development tools:** Visual Studio Code, Android Studio, Maven, Git, GitHub, GitHub Desktop, Microchip Studio, RStudio
- **AI-assisted development:** ChatGPT, Claude, GitHub Copilot; familiarity with LLM agents, tool calling and Model Context Protocol (MCP)
- **Engineering and design tools:** LTspice, Proteus, Altium Designer, AutoCAD
- **Creative tools:** Adobe Photoshop, Adobe Illustrator, Adobe InDesign, DaVinci Resolve
- **Collaboration:** Team-based development, code reviews, issue tracking and project coordination using GitHub, Trello and Slack

Additional Skills

- Analytical problem solving and attention to detail
- Technical documentation and report writing
- Technical presentations and clear communication
- Team collaboration and project coordination
- Staff training, teaching and volunteer coordination
- Organisation and willingness to learn unfamiliar tools and topics

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Work Experience

Senior Front of House
Sushi Chew Chew | 2024–Present

- Train and support new staff
- Complete opening and closing procedures
- Process customer orders and payments and complete end-of-day cash reconciliation
- Prepare food and beverages while supporting kitchen operations
- Monitor and replenish food and beverage stock

Holiday Programme Junior Volunteer → Senior Teacher
Rainbow Chinese Community Centre | 2018–2024 (Junior Volunteer); 2026–Present (Senior Teacher)

- Develop lesson plans and organise activities for children
- Lead classes and deliver lessons
- Coordinate and assign tasks to volunteers
- Supervise children and support their safety and participation
- Assist with food preparation, cleaning, and general programme operations

Front of House & Barista
Hollywood Bakery and Espresso / The Latte Cafe | 2020–2022

- Engaged with customers and responded to their individual needs
- Processed orders and payments and completed end-of-day cash reconciliation
- Prepared food and beverages while supporting kitchen operations

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Extracurricular Activities

Waikato Representative Badminton Athlete | 2018–2023

- Winner, Northern Region Boys' Local Derby, 2022
- Competed in the NZ U19 Junior Team Championships, 2022
- Competed in the Northern Region Boys' Local Derby, 2021
- Competed in the NZ U15 Team Championships, 2019

Sports and Recreation

- Participate in weekly social football, weightlifting, bouldering, and running

Social Media Engagement

- Managed social media engagement and grew an account from fewer than 100 to more than 1,000 followers within two months

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Right to work in New Zealand: New Zealand citizen

Driver licence: Full New Zealand driver licence

Linkedin | Facebook | GitHub